VIT
Vellore Institute of Technology

Reg. No. : 22BAI1266

Final Assessment Test (FAT) - JUNE/JULY 2023

Programme	B.Tech.	Semester	Winter Semester 2022-23
Course Title	STRUCTURED AND OBJECT-ORIENTED PROGRAMMING	Course Code	BCSE102L
Faculty Name	Prof. Sathian D	Slot	C2
		Class Nbr	CH2022232300568
Time	3 Hours	Max. Marks	100

PART A (7 X 10 Marks)
Answer All questions

01. Write a C program to suggest whether the family can select any health insurance policy for their entire family's well-being or not. This suggestion has been taken after considering the following factors and their weightage to calculate the health index of the family. [10]

Family Size – 10%
Number of Salaried People – 20%
Percentage of Adults in the family – 25%
Percentage of Older Adults in the family – 45%

Read the number of family members and their detail such as age and salary income status. If age is more than 55 then they are older adults. If the age is between 25 to 54 then they are adults. Calculate the percentage of adults and older adults in the family. Calculate the family health index by the weighted sum of all four above-mentioned factors. If the family health index is greater than 1.10 then the system has to give a 'Strong Recommendation' to take health insurance for the family. If the index is in the range of 0.75 to 1.09, then the system can recommend 'Advisable' or if the index is less than 0.75 then the system can recommend 'Not Necessary' to take the health insurance for the entire family.

02. Write a C program to manage employee records for a company. Each employee record should contain the following information: employee ID (integer), name (string), age (integer), salary (floating - point number), department (string), and position (string). Decide an appropriate data structure to store each employee detail in a collective manner. Decide on a mechanism to access multiple employee records. Write the C program with different modules to perform the following tasks: [10]

- Display the details of the employee with the highest Gross Salary.
- Display the details of the youngest employee.
- Calculate and display the average salary of all employees.
- Find and display the employees in a specific department.
- Find and display the employees holding a specific position.

03. Given a non-negative integer, write a C program to count the number of 1's present in the number using recursion. For example, if the given integer is 15,then the output should be 8 as there are eight 1's considering every digit of the numbers from 1 to 15. [10]

04. Develop a system for managing a university course enrolment. The system needs to handle different types of courses, including lectures and lab courses. Each course has details such as the course code, title, and instructor. Lecture courses have an additional detail called lecture hall, while lab courses have a lab number. Develop a C++ program that implements the following classes: *Course, LectureCourse*, and *LabCourse*. The *Course* class should serve as the base class for all types of courses and contain member variables *courseCode, title*, and *instructor*. The *LectureCourse* class should have a constructor that accepts the *course code, title, instructor*, and *lecture_hall*. The *LabCourse* class should have a constructor that accepts the course *code, title, instructor*, and *lab_number*. Implement member functions *displayDetails()* in each class that displays the details of the course. In your *main()* function, create instances of a lecture course and a lab course. Using their respective constructors, initialize their attributes, and display their details using the *displayDetails()* function. Name the inheritance type that is used in the above scenario. [10]

05. In the IPL final match between MI and CSK, there is an exciting moment when a batsman from the Mumbai Indians team hits a six against a bowler from the Chennai Super Kings team. The batsman is a part of the *MumbaiIndians* class, and the bowler belongs to the *ChennaiSuperKings* class. Use an appropriate inheritance concept to represent this scenario. Justify your answer. Use base class called *Player* which contains common attributes and methods for all players and derived classes for specific roles: *Batsman, Bowler* and *AllRounder*. There are derived classes specific to each team as well namely: *MumbaiIndians*, *ChennaiSuperKings* and *RoyalChallengersBangalore*. If these classes inherit from the appropriate role classes, identify the type of inheritance in that scenario. [10]

06. Write a C++ program to create a generic class for Subject with the member variables *Students' Identification* and *Value*. Assume that the number of registered students in the class is represented in the member variable count. The University web portal allows faculty to use any one of the two standard formats to maintain the details of the Subject. As the first method, the integer Roll Number can be used for *Students' Identification*, and the subject's Grade in character is used to represent the *Value*. In the Second format, the Student's name is for *Identification* and Mark is for *Value*. Write a program to create the objects under two given standards for the Student class. The class Student has *set_id* and *set_value* methods to initialize the data members. Also, it has a *put* method to display the names of the top 5 scorers in this subject. [10]

07. You are given a sorted array of integers 'Arr' which is in non-decreasing order and a target value 'X'. Display the value stored in the beginning of the array. Also display the position of the target element 'X'. If there are duplicated in the array for the target element, display only the first occurrence of the target element in the array. Write a C program for the above scenario and explain it with an example. [10]

PART B (2 X 15 Marks)

Answer All questions

08. The supermarket has two main categories of products: *Grocery* and *Electronics*. Under each category, there are further subcategories of products. For *Grocery*, *Food* and *Cleaning_Products* are the sub categories, and for *Electronics, Appliances* and *Gadgets* are the sub categories. There [15]

prints the grocery details available in that class. The *Electronics* class has a method called *GetInfo()* that gets the name of the electronic items from the admin user. Store those names in an array. There is a method called *DisplayInfo()* that prints the electronic items available in that class. Define the classes with their respective methods and inheritance relationships. Create objects for the derived classes and demonstrate the usage of the methods in the inheritance hierarchy. Implement the classes and write the necessary code to test the functionality.

09. Create a class called *BuyJewellery*. The cost of gold is Rs 5000 /-per gram and silver is Rs 80 /- and per gram of bronze is Rs 35 /-. Irrespective of the number of grams of purchase, if a customer buys gold, silver, and bronze as a combo he is given Rs 1000 /- flat discount on the net cost. If he buys only gold and bronze he is given Rs 750 as a discount and Rs 500 discount for silver and bronze combination, otherwise for any other combination there is no discount. Write a C++ program that has method named *discount()* to do all three categories of computations for every instance of purchase using an appropriate concept of object orientation. [15]